

Jack Yu

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EDUCATION

California Institute of Technology · Pasadena, CA Expected Jun. 2029
B.S. in Physics · GPA: 4.1 · Sophomore

Coursework: Quantum Mechanics, Linear Algebra, Abstract Algebra, Differential Equations, Statistical Mechanics, Multivariable Calculus, Classical Mechanics, Electromagnetism, Micro/Nanofabrication Lab

RESEARCH EXPERIENCE

Student Researcher — Experimental Gravity Group (EGG) · Caltech Jan 2026 – Present
Pasadena, CA

- Developed numerical methods to design non-Gaussian photonic quantum states for quantum sensing and computation using QuTiP and Strawberry Fields.
- Formulated multi-mode beam-splitter and squeezing circuits as high-dimensional parameter optimization problems, maximizing target-state fidelity under photon loss and phase noise.
- Implemented large-scale quantum state simulations and numerical optimization in Python, using NumPy, SciPy, TensorFlow, and HPC resources to parallelize parameter searches and fidelity calculations.
- Investigated how circuit architecture, squeezing parameters, loss, and measurement choices affect the achievable quality of generated quantum states.

William N. Lacey SURF Research Fellow — Cushing Lab · Caltech Summer 2026
Pasadena, CA

- Investigated microscopic mechanisms of ion transport in solid-state electrolytes through computational modeling and temperature-dependent spectroscopy.
- Modeled lattice anharmonicity in Na₃PS₄ using density functional theory (DFT) to study phonon-mediated ion transport.
- Designed and integrated a thermoelectric temperature-control system for Laser-driven Ultrafast Impedance Spectroscopy (LUIS).

Optoelectronics Intern · Sandia National Laboratories — III-V Photonics Center Jun 2024 – Sep 2025
Albuquerque, NM

- Conducted R&D for the DARPA H6 program developing a piezo-tunable Fabry-Pérot etalon for chip-scale atomic clocks.
- Characterized VCSEL devices through probe-station measurements and quantitative analysis of laser performance.

SELECTED PROJECT

Hexagonal Non-Gaussian State Generation · Python, QuTiP, Strawberry Fields 2026
Caltech EGG

- Designed parameterized multi-mode photonic circuits to generate a target hexagonal non-Gaussian quantum state.
- Optimized squeezing profiles, circuit parameters, and mode allocations to maximize state fidelity under realistic photon-loss models.
- Built numerical simulation and optimization workflows for high-dimensional quantum states and evaluated performance across circuit configurations.

PUBLICATION

D. K. Serkland, A. J. Grine, P.-S. Wong, **Jack Yu**. *Noise Characterization of VCSELs for Atomic Clocks and Magnetometers*. Proceedings of SPIE, Vol. PC13911, Vertical-Cavity Surface-Emitting Lasers XXX, 2026.

TECHNICAL SKILLS

Programming	Python, Java, MATLAB, Git, LaTeX
Quantum Computing	QuTiP, Strawberry Fields, Quantum State Simulation
Scientific Computing	NumPy, SciPy, TensorFlow, Numerical Optimization, DFT
Micro/Nanofabrication	Electron-Beam Lithography, Optical Lithography, AFM, SEM, Microfluidics
Experimental	Optical Alignment, Probe Station, PCB Design, Device Characterization

SELECTED AWARDS

- AIME Qualifier** — American Mathematics Invitational Examination
- NISCA All-American** — Top 100 in the US, 200 Medley Relay, Jun 2025

EXTRACURRICULARS

NCAA Division III Varsity Swim Team · Caltech 2025 – Present
Chamber Music Ensemble (Piano) · Caltech Present